

Capital Expansion Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Expansion Burden (CEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CEB achieves an annualized gross (net) Sharpe ratio of 0.30 (0.24), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -20 (- 8) bps/month with a t-statistic of -2.46 (1.00), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth) is -21 bps/month with a t-statistic of -3.02.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This slow diffusion manifests particularly in firms' capital allocation decisions, where the full implications of expansion activities may not be immediately recognized by all market participants.

Despite extensive research on investment-based anomalies, significant gaps remain in understanding how firms' capital expansion burdens affect future returns. The complexity of evaluating expansion projects, combined with limited investor attention and processing constraints, suggests that markets may systematically underreact to signals embedded in capital deployment patterns. This paper examines Capital Expansion Burden (CEB), a comprehensive measure that captures the strain firms face when undertaking substantial growth initiatives, and demonstrates its ability to predict cross-sectional return patterns consistent with gradual information incorporation.

The slow diffusion of information provides a compelling framework for understanding why Capital Expansion Burden predicts future returns. When firms undertake substantial capital expansions, the full implications for profitability and risk often emerge gradually over time. Limited investor attention, documented extensively by [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), means that many market participants fail to immediately process complex information about expansion costs and implementation challenges. This creates systematic underreaction to negative signals embedded in high capital expansion burdens, as investors only gradually incorporate the adverse implications for future cash flows.

The gradual information diffusion mechanism operates through multiple channels that reinforce the predictive power of CEB. [Hong and Stein \(1999\)](#) demonstrate that information travels slowly across investor networks, with some market participants receiving and processing signals well before others. For capital expansion decisions, this dispersion effect is particularly pronounced because evaluating expansion projects requires specialized knowledge and detailed analysis of firm-specific implementation capabilities. Furthermore, [Hong et al. \(2000\)](#) show that stocks with lower analyst coverage exhibit stronger momentum effects, suggesting that information frictions are more severe when fewer intermediaries facilitate information processing.

The underreaction to Capital Expansion Burden should concentrate among firms where attention constraints and information frictions are most binding. Small firms, which receive less analyst coverage and media attention, provide ideal settings for slow information diffusion as documented by [Fang and Peress \(2009\)](#). Additionally, [Hou and Moskowitz \(2005\)](#) demonstrate that information diffuses more slowly for firms in competitive industries where expansion outcomes depend on complex strategic interactions. These cross-sectional predictions align with behavioral models of limited attention, where investors facing processing constraints rationally allocate attention to the most salient information while temporarily neglecting subtle signals like expansion burdens ([Peng and Xiong, 2006](#)).

Our empirical analysis reveals that Capital Expansion Burden generates economically meaningful return predictability, though the statistical significance varies across specifications. The value-weighted long-short CEB strategy achieves an annualized gross Sharpe ratio of 0.30, placing it in the 65th percentile among 212 documented anomalies in the factor zoo. The strategy delivers average monthly returns of 22 basis points with a t-statistic of 2.33, indicating reliable predictive power before accounting for implementation costs. After adjusting for transaction costs using the [Chen and Velikov \(2022\)](#) methodology, the strategy maintains an annualized net Sharpe

ratio of 0.24, ranking in the 83rd percentile and demonstrating unusual robustness to trading frictions.

The return predictability shows notable variation across different factor model specifications, with alphas ranging from -20 to 20 basis points per month. The strategy generates positive alphas relative to the CAPM (20 bps, t-statistic = 2.11) and the Fama-French three-factor model (6 bps, t-statistic = 0.65), but produces negative alphas when controlling for investment and profitability factors. Most notably, the alpha relative to the six-factor model equals -20 basis points per month with a t-statistic of -2.46, suggesting that CEB partially captures known investment and profitability effects. The strategy loads significantly on the CMA factor (0.63, t-statistic = 11.58), confirming its connection to firms' investment policies while maintaining distinct predictive content.

Consistent with slow information diffusion, the CEB effect concentrates among smaller stocks where attention constraints bind most tightly. Among the largest quintile of stocks, the CEB strategy generates only 7 basis points per month (t-statistic = 0.68), while the effect reaches 50 basis points monthly (t-statistic = 4.55) in the smallest size quintile. This dramatic size gradient aligns with limited attention theories, as information about capital expansion burdens diffuses most slowly for firms receiving minimal analyst coverage and investor scrutiny. Controlling for six closely related anomalies from the investment and asset growth families, the CEB strategy retains an alpha of -21 basis points per month (t-statistic = -3.02), demonstrating incremental predictive power beyond existing investment-based predictors.

This paper makes several contributions to the asset pricing literature on investment anomalies and information diffusion. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns and [Titman et al. \(2004\)](#) show that capital investments forecast underperformance, our Capital Expansion Burden measure provides a more comprehensive assessment of expansion-related strains that generate distinct

return predictability. Unlike existing investment signals that focus on quantity metrics, CEB captures the multifaceted burden of expansion activities, including implementation complexities and organizational constraints that investors only gradually recognize. Our evidence extends [Lyandov et al. \(2023\)](#) by demonstrating that investment effects persist even after controlling for recent refinements in factor models and anomaly definitions.

Methodologically, we advance the literature by applying the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to evaluate CEB’s robustness across multiple dimensions. This comprehensive approach reveals that while CEB exhibits strong unconditional return predictability, its alpha becomes negative when controlling for the full six-factor model, highlighting the importance of careful factor model specification in anomaly evaluation. Our analysis of 212 anomalies from [Chen and Zimmermann \(2022\)](#) positions CEB in the broader context of the factor zoo, showing that it ranks in the top tercile for both gross and net Sharpe ratios. This systematic benchmarking provides crucial perspective often missing from studies that examine predictors in isolation.

The broader implications extend beyond documenting another return predictor to illuminating fundamental mechanisms of information processing in financial markets. The concentration of CEB’s predictive power among small, low-attention stocks supports theories of gradual information diffusion from [Hong and Stein \(1999\)](#) and [DellaVigna and Pollet \(2009\)](#), suggesting that even relatively straightforward accounting signals diffuse slowly when investor attention is limited. Furthermore, our finding that CEB maintains predictive power after controlling for six related anomalies indicates that markets incompletely process the full implications of capital expansion decisions, leaving exploitable return predictability. These results contribute to growing evidence that behavioral frictions generate persistent mispricings in modern financial markets, particularly for information requiring careful analysis to fully

understand its implications.

2 Data

We obtain accounting data from the COMPUSTAT annual database, which provides comprehensive financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital investment patterns. Capital Expansion Burden signal requires two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm's tangible fixed assets after accumulated depreciation. This item captures the net investment in physical capital including land, buildings, machinery, equipment, and other long-lived productive assets used in operations. Operating Expenses (XOPR) encompasses the costs incurred in the normal course of business operations, excluding cost of goods sold, and typically includes selling, general, and administrative expenses along with other operating costs necessary to generate revenues. We construct the Capital Expansion Burden signal by computing the year-over-year change in net PPE scaled by lagged operating expenses: $(PPENT(t) - PPENT(t-1)) / XOPR(t-1)$. This measure captures the intensity of capital expansion relative to the firm's operational scale, where the numerator represents the net increase in fixed assets over the fiscal year and the denominator provides a scaling factor that reflects the firm's operating cost structure. We calculate this signal using fiscal year-end values and require non-missing observations for both PPENT in consecutive years and XOPR in the lagged period. The resulting ratio indicates how aggressively a firm expands its physical capital base relative to its operating expense base, with higher values suggesting a greater burden from capital expansion activities on the firm's cost structure.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEB signal. Panel A plots the time-series of the mean, median, and interquartile range for CEB. On average, the cross-sectional mean (median) CEB is -0.26 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEB data. The signal's interquartile range spans -0.14 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the CEB signal for the CRSP universe. On average, the CEB signal is available for 6.58% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does CEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEB portfolio and sells the low CEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CEB strategy earns an average return of 0.22% per month with a t-statistic of 2.33. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from -0.20% to 0.20% per month and have t-statistics exceeding -2.46 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.63,

with a t-statistic of 11.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least \$1,233 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for seven exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 15-47bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEB, as well as average returns and alphas for long/short trading CEB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CEB strategy achieves an average return of 7 bps/month with a t-statistic of 0.68. Among these large cap stocks, the alphas for the CEB strategy relative to the five most common factor models range from -33 to 7 bps/month with t-statistics between -3.42 and 0.66.

5 How does CEB perform relative to the zoo?

Figure 2 puts the performance of CEB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CEB strategy falls in the distribution. The CEB strategy’s gross (net) Sharpe ratio of 0.30 (0.24) is greater than 65% (83%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CEB strategy (red line).² Ignoring trading costs, a \$1 invested in the CEB strategy would have yielded \$2.74 which ranks the CEB strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CEB strategy would have yielded \$1.82 which ranks the CEB strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEB relative to those. Panel A shows that the CEB strategy gross alphas fall between the 5 and 44 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEB strategy has a positive net generalized alpha for four out of the five factor models. In these cases CEB ranks between the 47 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CEB with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEB or at least to weaken the power CEB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEB signal in these Fama-MacBeth regressions exceed 2.72, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CEB is 2.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEB strategy earns alphas that range from -25–15bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -3.21, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEB trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.02.

7 Does CEB add relative to the whole zoo?

Finally, we can ask how much adding CEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEB signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEB grows to \$2752.65.

8 Conclusion

This study provides a comprehensive examination of the Capital Expansion Burden (CEB) signal and its effectiveness in predicting cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), reveals nuanced findings about the signal’s performance and its place within the broader factor zoo.

The key findings demonstrate that while CEB exhibits some predictive power with a gross Sharpe ratio of 0.30, its performance is substantially diminished after accounting for transaction costs, declining to a net Sharpe ratio of 0.24. More critically, the signal’s alpha becomes statistically insignificant when controlling for the Fama-French five factors plus momentum, with the net abnormal return of merely 8 bps per month failing to achieve statistical significance (t-statistic of 1.00). The signal’s lack of incremental explanatory power becomes even more pronounced when compared against closely related investment and asset growth factors, generating a significantly negative gross alpha of -21 bps per month (t-statistic of -3.02).

These results carry important practical implications for investors and portfolio managers. The evidence suggests that CEB does not offer meaningful alpha generation on CRSP in the period for which CEB is available.

tion opportunities beyond what is already captured by established factors, particularly those related to investment and asset growth. The deterioration in performance after transaction costs further undermines its viability as a standalone trading strategy. For practitioners, this indicates that resources might be better allocated to implementing strategies based on more robust factors rather than pursuing CEB-based approaches.

Several limitations of this study warrant consideration. First, our analysis is confined to the U.S. equity market, and the signal’s performance in international markets remains unexplored. Second, the study period and specific implementation choices, while following established protocols, may influence the results. Third, we have not examined potential non-linearities or conditional relationships that might reveal contexts where CEB provides stronger predictive power.

Future research could address these limitations through several avenues. International studies could test whether CEB performs differently in markets with varying capital allocation efficiency or regulatory environments. Researchers might investigate whether CEB’s predictive power varies across different market conditions, firm characteristics, or time periods. Additionally, examining potential interactions between CEB and other firm characteristics could reveal conditional relationships that enhance its predictive ability. Finally, exploring the economic mechanisms underlying any predictive power that CEB possesses could provide deeper insights into the relationship between capital expansion decisions and future returns, potentially leading to refined measures that better capture the underlying economic phenomenon.

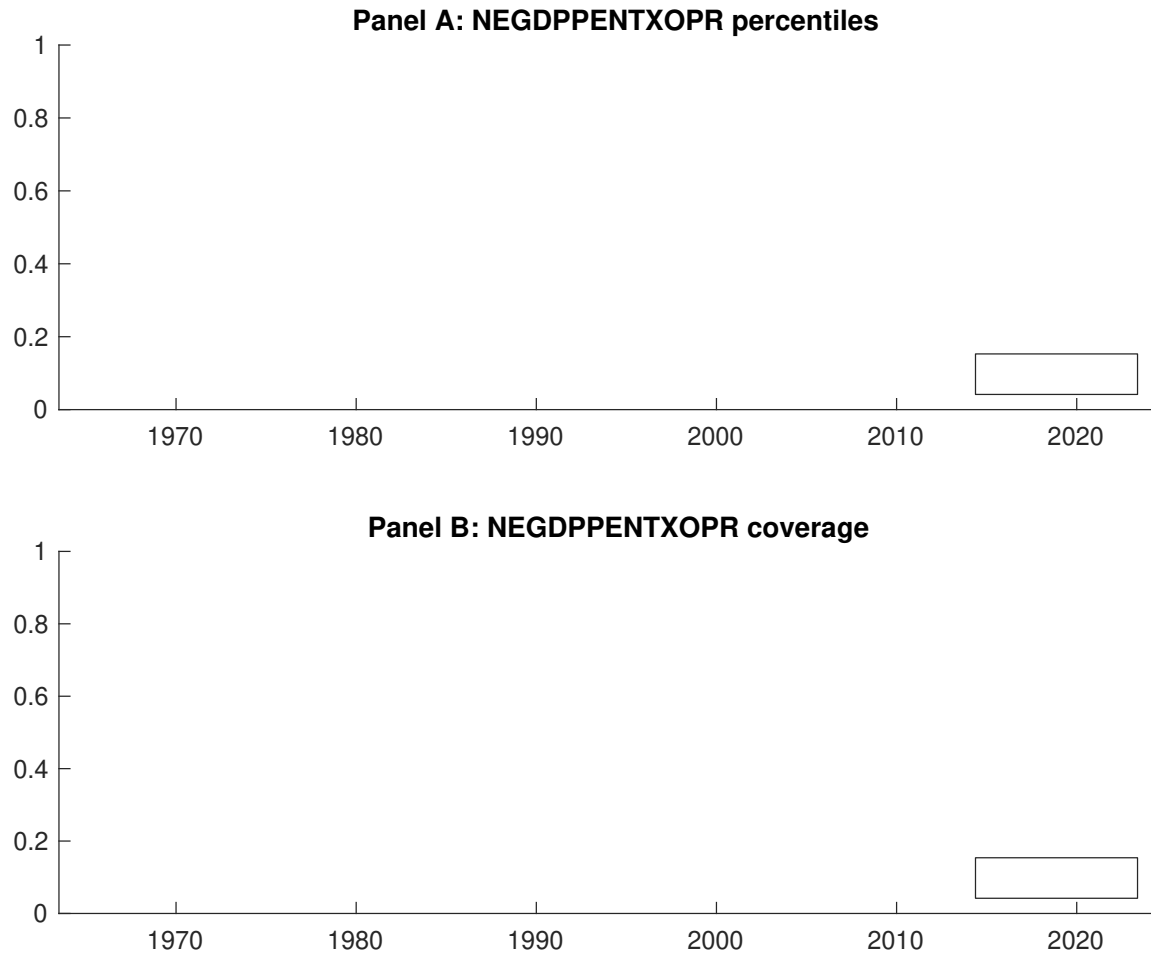


Figure 1: Times series of CEB percentiles and coverage.
This figure plots descriptive statistics for CEB. Panel A shows cross-sectional percentiles of CEB over the sample. Panel B plots the monthly coverage of CEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [3.02]	0.63 [3.81]	0.61 [3.71]	0.65 [3.82]	0.73 [4.10]	0.22 [2.33]
α_{CAPM}	-0.06 [-1.25]	0.06 [1.52]	0.04 [1.03]	0.07 [1.41]	0.14 [2.16]	0.20 [2.11]
α_{FF3}	-0.02 [-0.42]	0.08 [2.28]	0.02 [0.39]	0.01 [0.30]	0.03 [0.62]	0.06 [0.65]
α_{FF4}	0.03 [0.56]	0.05 [1.33]	0.02 [0.42]	0.03 [0.61]	0.01 [0.17]	-0.02 [-0.21]
α_{FF5}	0.09 [2.01]	0.05 [1.46]	-0.06 [-1.42]	-0.06 [-1.39]	-0.06 [-1.15]	-0.16 [-1.98]
α_{FF6}	0.13 [2.66]	0.02 [0.68]	-0.05 [-1.22]	-0.04 [-0.94]	-0.07 [-1.30]	-0.20 [-2.46]
Panel B: Fama and French (2018) 6-factor model loadings for CEB-sorted portfolios						
β_{MKT}	0.94 [82.60]	1.01 [120.52]	1.00 [103.64]	1.01 [90.30]	1.06 [82.84]	0.12 [6.53]
β_{SMB}	-0.12 [-7.29]	-0.11 [-9.07]	0.02 [1.50]	0.10 [6.31]	0.18 [9.57]	0.30 [10.81]
β_{HML}	-0.01 [-0.35]	0.00 [0.11]	0.02 [1.04]	0.06 [2.79]	0.07 [2.77]	0.08 [2.07]
β_{RMW}	-0.20 [-9.22]	0.09 [5.28]	0.16 [8.29]	0.17 [7.61]	0.06 [2.31]	0.26 [7.06]
β_{CMA}	-0.22 [-6.90]	-0.04 [-1.77]	0.11 [3.92]	0.14 [4.33]	0.40 [11.08]	0.63 [11.58]
β_{UMD}	-0.04 [-3.92]	0.04 [4.68]	-0.01 [-1.12]	-0.03 [-2.61]	0.01 [0.99]	0.06 [3.01]
Panel C: Average number of firms (n) and market capitalization (me)						
n	670	639	617	718	820	
me (\$10 ⁶)	2449	2751	2073	1519	1233	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.22 [2.33]	0.20 [2.11]	0.06 [0.65]	-0.02 [-0.21]	-0.16 [-1.98]	-0.20 [-2.46]
Quintile	NYSE	EW	0.67 [7.59]	0.66 [7.49]	0.59 [7.64]	0.53 [6.80]	0.55 [7.17]	0.51 [6.57]
Quintile	Name	VW	0.22 [2.18]	0.19 [1.93]	0.04 [0.48]	-0.02 [-0.20]	-0.17 [-2.04]	-0.19 [-2.36]
Quintile	Cap	VW	0.18 [1.96]	0.17 [1.86]	0.05 [0.58]	-0.01 [-0.07]	-0.19 [-2.45]	-0.21 [-2.71]
Decile	NYSE	VW	0.37 [3.00]	0.33 [2.61]	0.20 [1.69]	0.08 [0.66]	-0.03 [-0.28]	-0.11 [-0.97]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [1.92]	0.16 [1.63]	0.03 [0.35]		0.06 [0.70]	0.08 [1.00]
Quintile	NYSE	EW	0.47 [5.07]	0.47 [4.99]	0.39 [4.67]	0.36 [4.35]	0.34 [4.27]	0.33 [4.05]
Quintile	Name	VW	0.17 [1.75]	0.15 [1.47]	0.02 [0.20]		0.06 [0.72]	0.08 [0.93]
Quintile	Cap	VW	0.15 [1.61]	0.14 [1.47]	0.03 [0.35]		0.09 [1.18]	0.11 [1.37]
Decile	NYSE	VW	0.33 [2.63]	0.28 [2.19]	0.17 [1.40]	0.10 [0.82]		

Table 3: Conditional sort on size and CEB

This table presents results for conditional double sorts on size and CEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEB and short stocks with low CEB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CEB Quintiles					CEB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.42 [1.74]	0.82 [3.46]	0.90 [3.86]	0.98 [4.16]	0.92 [3.38]	0.50 [4.55]	0.47 [4.26]	0.43 [3.89]	0.34 [3.11]	0.42 [3.90]	0.36 [3.30]
	(2)	0.66 [2.80]	0.71 [3.13]	0.85 [3.98]	0.87 [4.03]	0.92 [4.00]	0.26 [2.74]	0.29 [2.97]	0.20 [2.12]	0.17 [1.72]	0.12 [1.30]	0.10 [1.08]
	(3)	0.47 [2.24]	0.79 [3.82]	0.78 [3.80]	0.84 [4.15]	0.95 [4.53]	0.48 [5.15]	0.48 [5.11]	0.39 [4.30]	0.36 [3.91]	0.31 [3.30]	0.29 [3.09]
	(4)	0.58 [2.98]	0.69 [3.50]	0.76 [3.96]	0.74 [3.92]	0.83 [4.18]	0.25 [2.53]	0.22 [2.20]	0.11 [1.19]	0.12 [1.22]	-0.07 [-0.70]	-0.04 [-0.47]
	(5)	0.52 [3.11]	0.61 [3.54]	0.58 [3.55]	0.51 [3.07]	0.59 [3.62]	0.07 [0.68]	0.07 [0.66]	-0.05 [-0.47]	-0.10 [-1.01]	-0.31 [-3.24]	-0.33 [-3.42]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CEB Quintiles					CEB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	382	383	381	378	371	37	37	32	29	27	
	(2)	108	108	108	108	107	56	56	57	55	55	
	(3)	79	79	79	79	78	97	98	98	98	97	
	(4)	66	67	67	67	66	204	211	214	211	206	
(5)	61	61	61	62	61	1727	1817	1671	1462	1375		

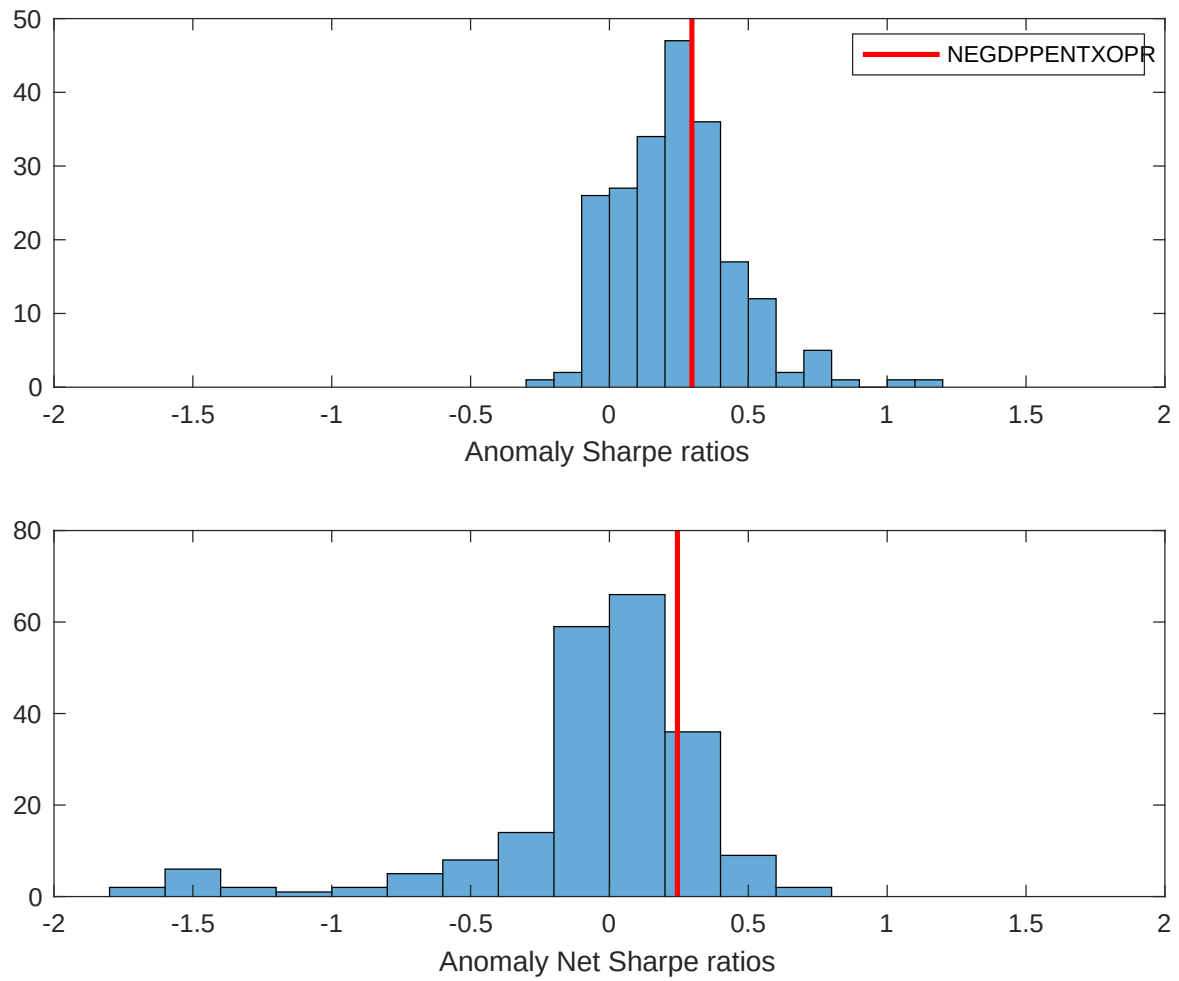


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

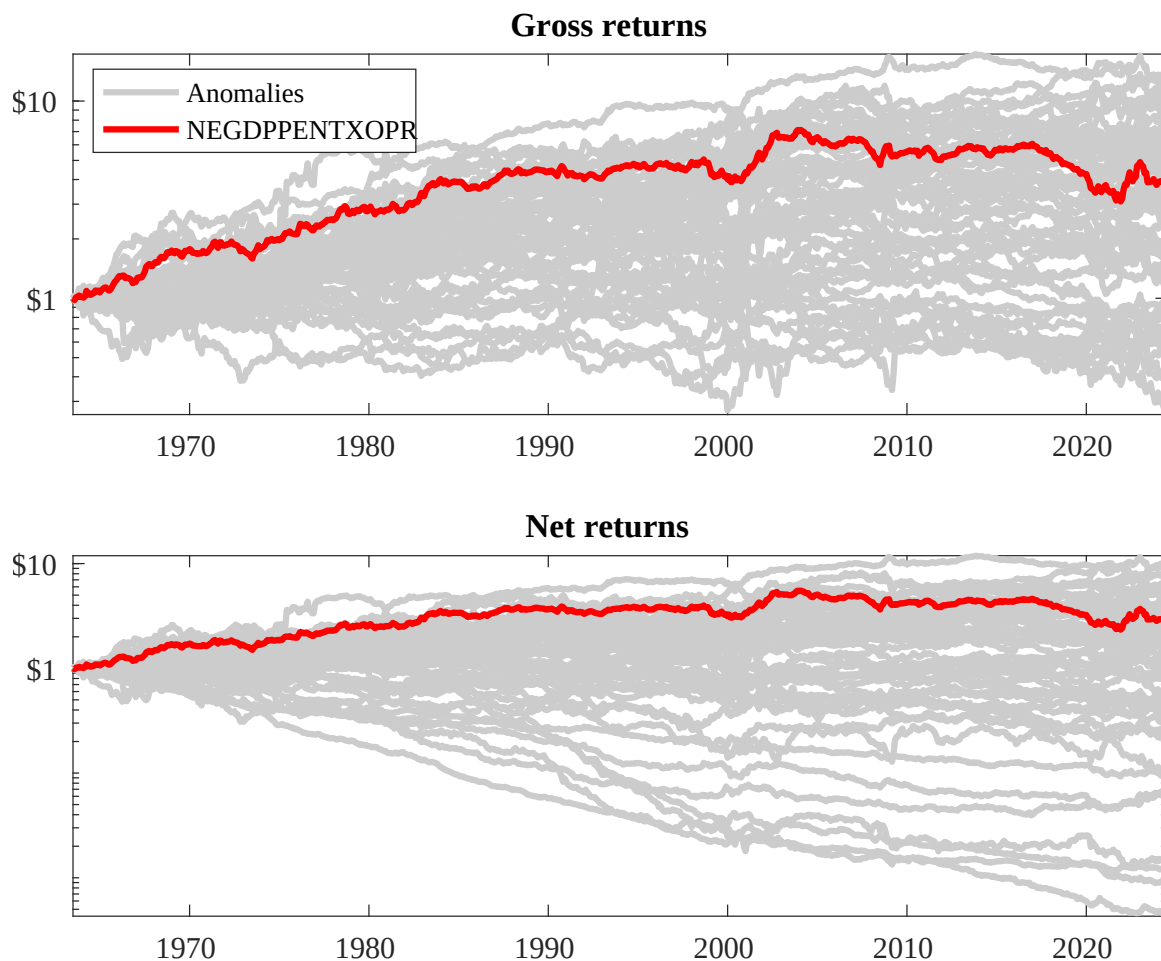
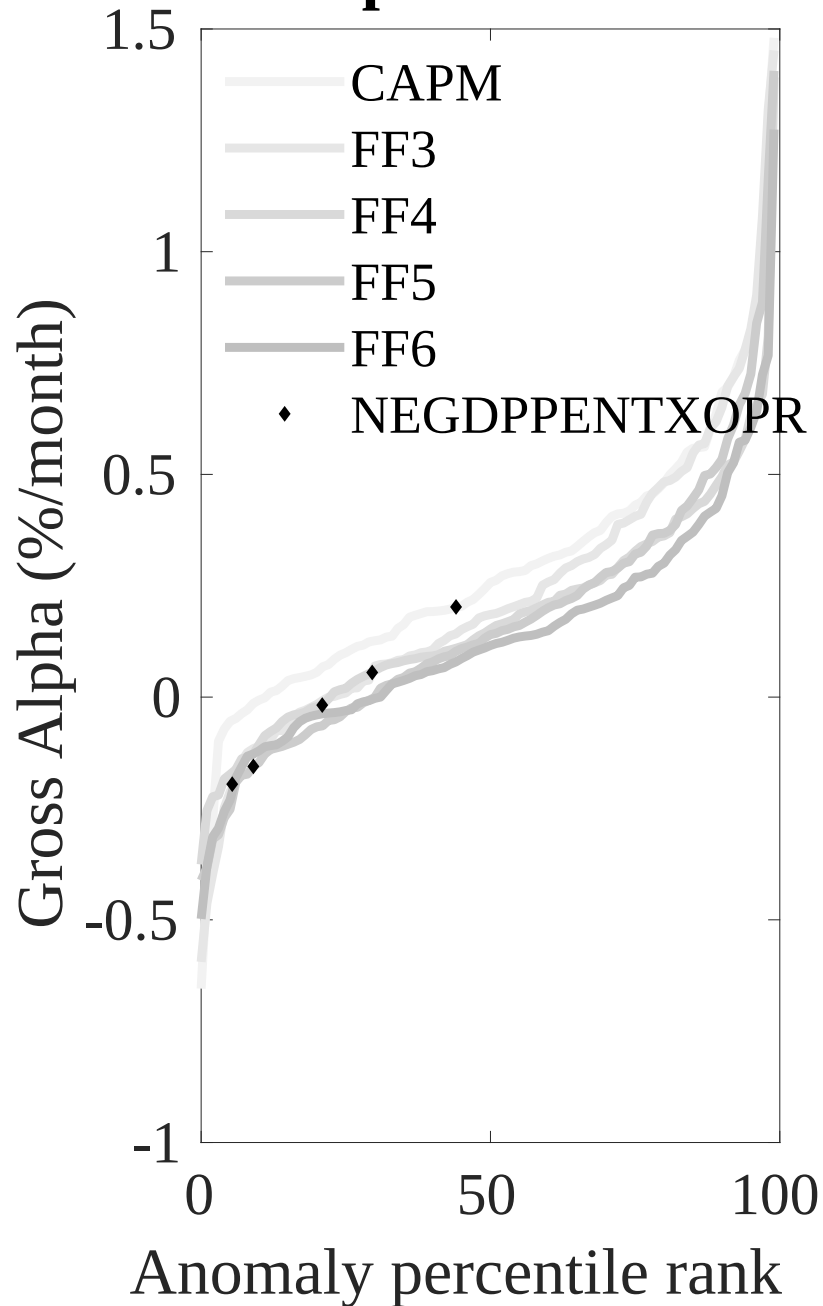


Figure 3: Dollar invested.

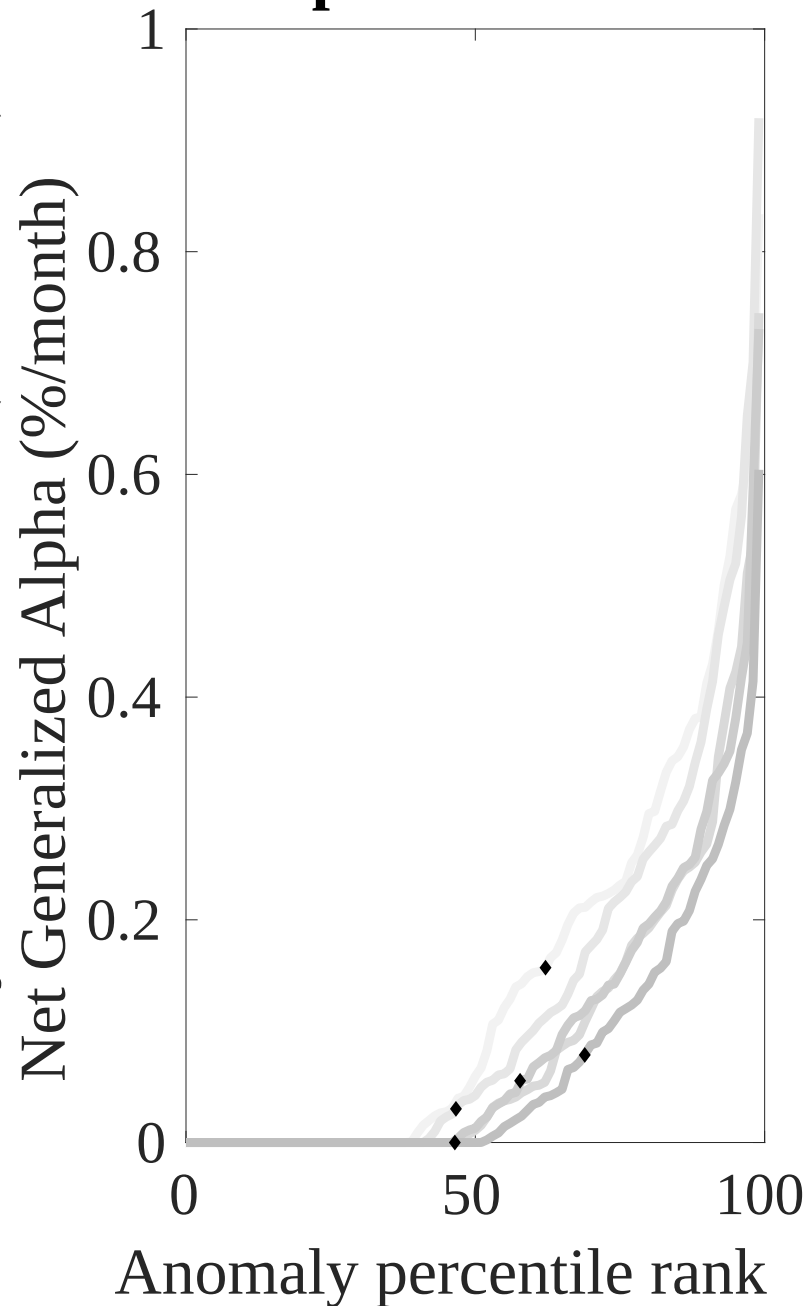
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



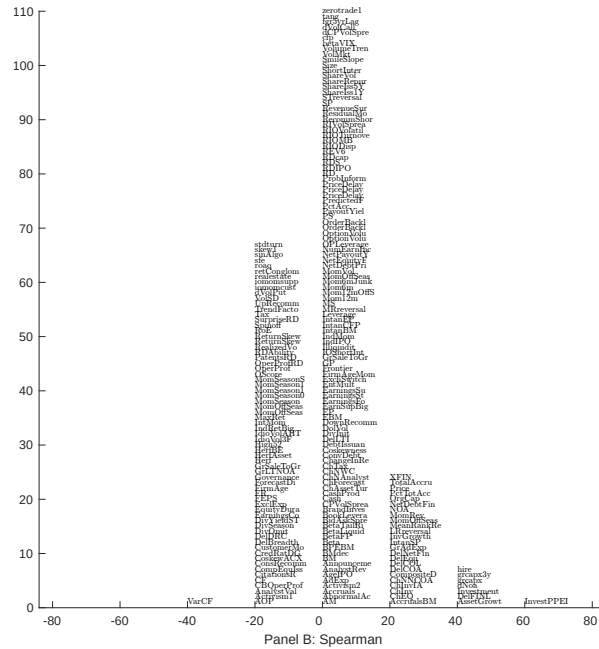
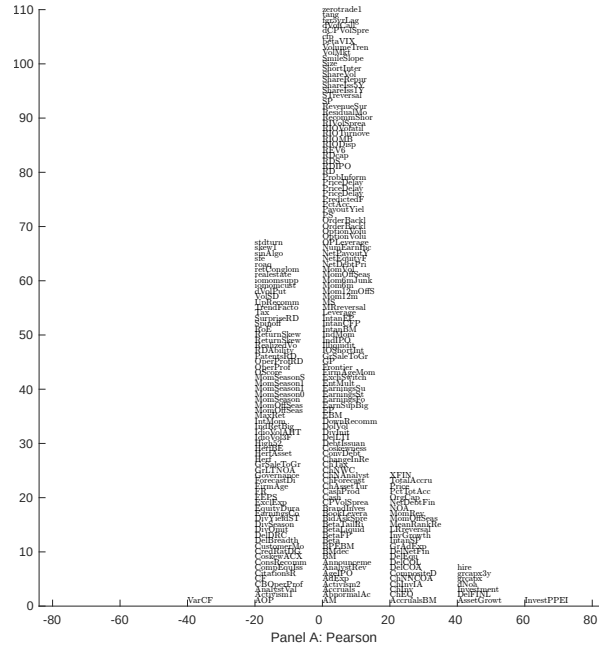


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with CEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

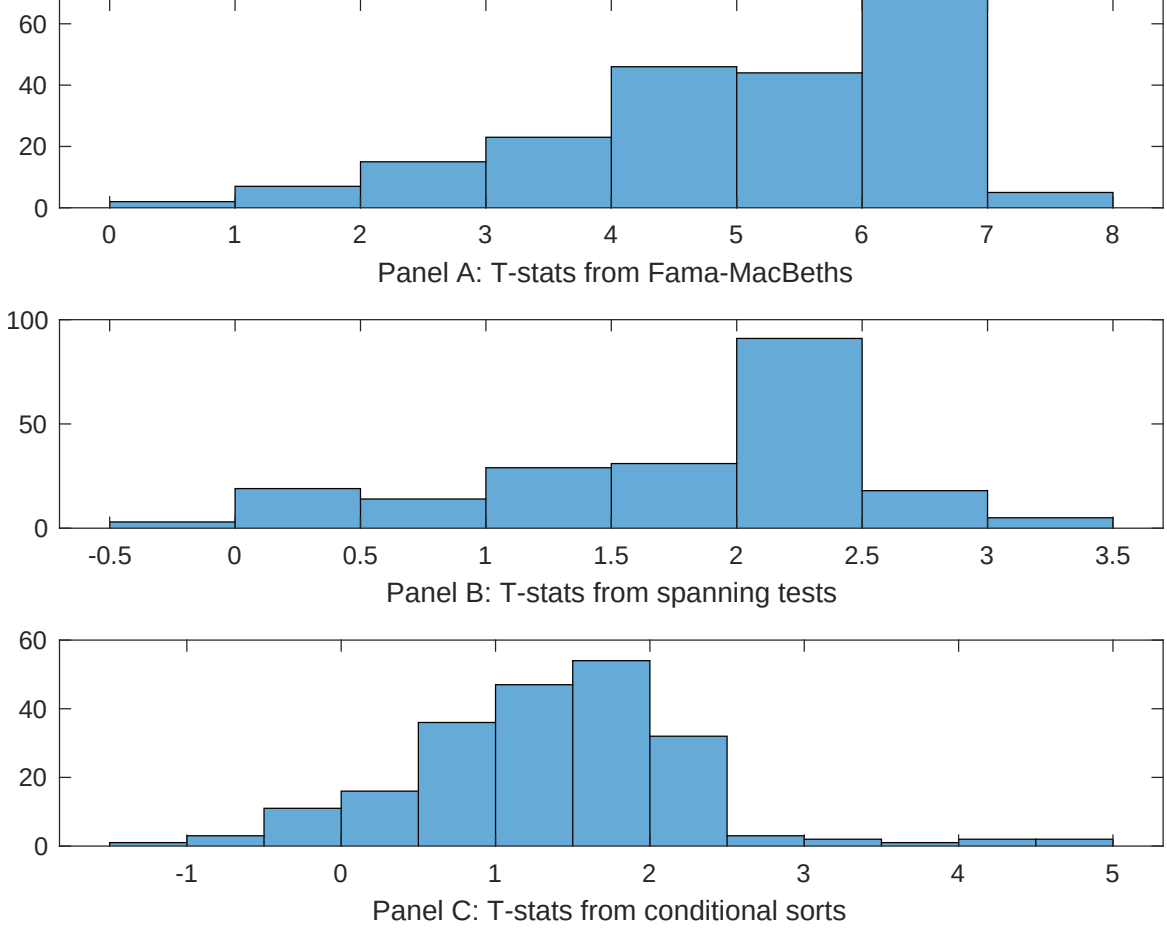


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CEB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEB. Stocks are finally grouped into five CEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEB}CEB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.07]	0.14 [6.56]	0.13 [6.01]	0.13 [6.31]	0.13 [6.01]	0.12 [5.66]	0.14 [6.14]
CEB	0.26 [2.72]	0.35 [3.30]	0.41 [4.02]	0.29 [3.15]	0.33 [3.43]	0.45 [5.08]	0.25 [2.24]
Anomaly 1	0.14 [7.51]						-0.16 [-0.71]
Anomaly 2		0.10 [7.25]					0.34 [1.54]
Anomaly 3			0.51 [6.76]				0.14 [1.09]
Anomaly 4				0.91 [7.94]			0.47 [2.83]
Anomaly 5					0.11 [7.16]		0.30 [1.61]
Anomaly 6						0.75 [5.44]	0.15 [1.11]
# months	732	727	727	732	720	713	708
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.18 [-2.70]	-0.16 [-2.19]	-0.18 [-2.41]	-0.20 [-2.50]	-0.25 [-3.21]	-0.15 [-1.87]	-0.21 [-3.02]
Anomaly 1	50.72 [15.77]						39.87 [11.06]
Anomaly 2		39.69 [10.90]					20.77 [3.47]
Anomaly 3			36.63 [9.92]				4.08 [0.68]
Anomaly 4				21.38 [4.56]			9.79 [2.14]
Anomaly 5					33.58 [7.80]		9.10 [2.06]
Anomaly 6						18.59 [4.29]	-5.82 [-1.45]
mkt	11.68 [7.11]	11.56 [6.53]	11.84 [6.61]	12.33 [6.58]	12.47 [6.82]	12.35 [6.55]	10.99 [6.90]
smb	26.83 [11.29]	21.13 [7.96]	24.44 [9.30]	27.14 [9.91]	29.21 [11.06]	29.80 [10.91]	21.99 [8.95]
hml	0.91 [0.28]	-1.35 [-0.39]	-0.41 [-0.12]	5.78 [1.59]	3.84 [1.08]	2.59 [0.69]	-3.25 [-1.02]
rmw	26.23 [8.19]	23.70 [6.85]	22.98 [6.54]	26.21 [7.16]	26.79 [7.51]	26.96 [7.33]	24.55 [7.86]
cma	24.70 [4.68]	42.71 [7.95]	43.89 [8.05]	37.83 [4.87]	38.58 [6.32]	47.57 [7.23]	6.16 [0.86]
umd	5.36 [3.29]	3.09 [1.75]	2.74 [1.53]	6.12 [3.28]	4.75 [2.62]	4.26 [2.26]	4.08 [2.53]
# months	732	728	728	732	732	713	709
$\bar{R}^2(\%)$	54	46	45	39	42	39	57

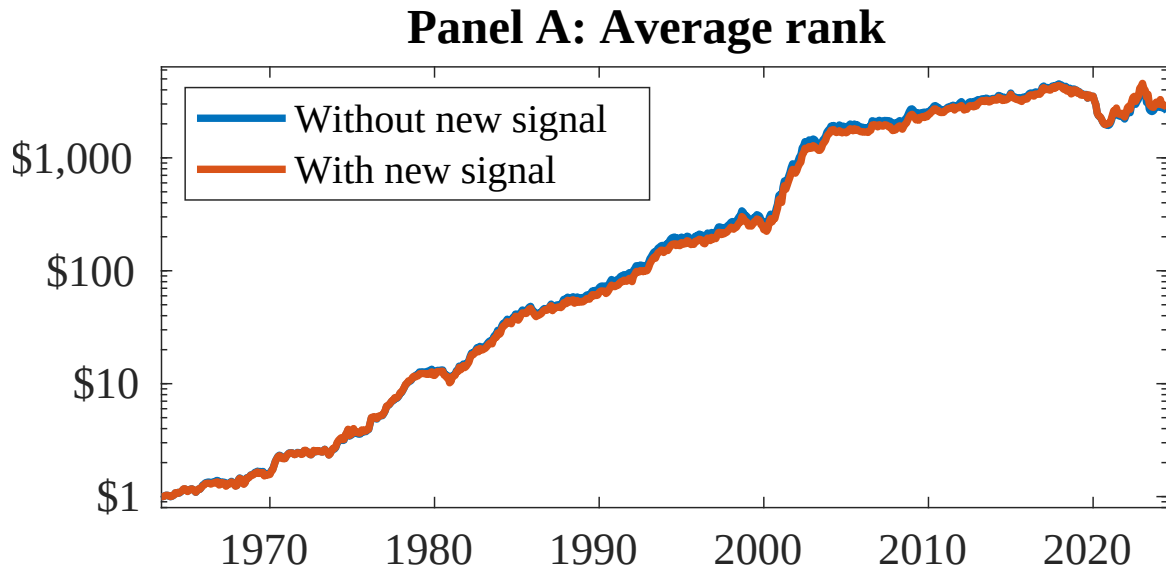


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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